

Shell Topic 07: sed

Note 1. The `sed` (*stream editor*) edits files line-by-line. The most common `sed` command is *substitution* `s/REGEX/REPLACEMENT/FLAGS`. By default, substitution replaces only the first match on each line. The `g` (global) flag replaces every match on each line. The `/` in `s/a/b/` is not special. `Sed` uses whatever character immediately follows the `s` as the delimiter. Therefore `s;a;b;`, `s|a|b|`, and `s,a,b,` are all equivalent to `s/a/b/`.

Problem 2. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > README <<EOF
3 hello hello hello
4 hola hola mundo
5 salve munde
6 EOF
7 $ cat README | sed 's/hello/goodbye/' | grep 'hello' | wc -l
```

Fraction of LLMs with correct answer: 8 / 13 = 0.62

Problem 3. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > README <<EOF
3 hello hello hello
4 hola hola mundo
5 salve munde
6 EOF
7 $ cat README | sed 's/hello/goodbye/g' | grep 'hello' | wc -l
```

Fraction of LLMs with correct answer: 13 / 13 = 1.00

Problem 4. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > README <<EOF
3 hello hello hello
4 hola hola mundo
5 salve munde
6 EOF
7 $ cat README | sed 's/hola/hello/' | sed 's/hello/goodbye/g' | grep 'hola' | wc -l
```

Fraction of LLMs with correct answer: 3 / 13 = 0.23

Problem 5. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > paths <<EOF
3 /usr/local/bin/python3
4 /usr/local/lib/python3
5 /usr/bin/sed
6 EOF
7 $ cat paths | sed 's/\/usr\/local\/\/opt/' | grep '/opt' | wc -l
```

Fraction of LLMs with correct answer: $12 / 13 = 0.92$

Problem 6. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > paths <<EOF
3 /usr/local/bin:/usr/local/lib
4 /usr/bin:/usr/lib
5 EOF
6 $ cat paths | sed 's;/usr/local;/opt;' | grep '/usr/local' | wc -l
```

Fraction of LLMs with correct answer: $5 / 13 = 0.38$

Problem 7. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > paths <<EOF
3 /usr/local/bin:/usr/local/lib
4 /usr/bin:/usr/lib
5 EOF
6 $ cat paths | sed 's;/usr/local;/opt;g' | grep '/usr/local' | wc -l
```

Fraction of LLMs with correct answer: $11 / 13 = 0.85$

Note 8. `sed` scripts are ordinary shell words. This creates a notorious source of bugs because the characters `$` and `^` mean different things to the shell and to `sed`. To `sed`, `^` matches the beginning of a line and `$` matches the end of a line. To the shell, `$` is the variable expansion operator. The shell expands variables inside double quotes `"` but not inside single quotes `'`. Recall that the shell does not expand the variable expansion operator `$` when it is preceded by a backslash `\` or is not followed by a letter/number.

Problem 9. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ place=world
3 $ cat > README <<EOF
4 hello world
5 hola mundo
6 salve munde
7 EOF
8 $ cat README | sed "s/$place/mundo/" | grep 'mundo' | wc -l
```

Fraction of LLMs with correct answer: $9 / 13 = 0.69$

Problem 10. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ place=world
3 $ cat > README <<EOF
4 hello world
5 hola mundo
6 salve munde
7 EOF
8 $ cat README | sed 's/$place/mundo/' | grep 'mundo' | wc -l
```

Fraction of LLMs with correct answer: $9 / 13 = 0.69$

Problem 11. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > README <<EOF
3 hello world
4 world hello
5 salve munde
6 EOF
7 $ cat README | sed 's/world$/mundo/' | grep 'mundo' | wc -l
```

Fraction of LLMs with correct answer: $9 / 13 = 0.69$

Problem 12. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ world=munde
3 $ cat > README <<EOF
4 hello world
5 world hello
6 salve munde
7 EOF
8 $ cat README | sed "s/world$/mundo/" | grep 'mundo' | wc -l
```

Fraction of LLMs with correct answer: $6 / 13 = 0.46$

Problem 13. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ greeting=hello
3 $ cat > README <<EOF
4 hello world
5 world hello
6 salve munde
7 EOF
8 $ cat README | sed "s/^$greeting/hola/" | grep 'hola' | wc -l
```

Fraction of LLMs with correct answer: $10 / 13 = 0.77$

Problem 14. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ greeting=hello
3 $ cat > README <<EOF
4 hello world
5 world hello
6 salve munde
7 EOF
8 $ cat README | sed 's/^$greeting/hola/' | grep 'hola' | wc -l
```

Fraction of LLMs with correct answer: $8 / 13 = 0.62$

Problem 15. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ place=world
3 $ cat > README <<EOF
4 hello $place
5 hola $place
6 salve munde
7 EOF
8 $ cat README | sed 's/world/mundo/' | grep 'mundo' | wc -l
```

Fraction of LLMs with correct answer: $7 / 13 = 0.54$

Problem 16. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ place=world
3 $ cat > README <<'EOF'
4 hello $place
5 hola $place
6 salve munde
7 EOF
8 $ cat README | sed 's/world/mundo/' | grep 'mundo' | wc -l
```

Fraction of LLMs with correct answer: $5 / 13 = 0.38$

Problem 17. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ place=world
3 $ cat > README <<'EOF'
4 hello $place
5 hola $place
6 salve munde
7 EOF
8 $ cat README | sed 's/$place/mundo/' | grep 'mundo' | wc -l
```

Fraction of LLMs with correct answer: $3 / 13 = 0.23$

LLM Model Performance

